

Dynamics of a Two-Level Atom Driven by a Classical Electromagnetic Field

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Abstract

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The interaction between a two-level atom and a monochromatic classical electromagnetic field is one of the fundamental models of quantum optics. Despite its apparent simplicity, it describes essential phenomena such as Rabi oscillations, resonance, and the geometric evolution of an atomic quantum state on the Bloch sphere.

In the semiclassical description, the atom is treated quantum mechanically while the driving field is treated classically. The resulting Hamiltonian is explicitly time dependent and includes both co-rotating and counter-rotating interaction terms. Under weak-coupling and near-resonance conditions, the counter-rotating terms can be neglected through the rotating-wave approximation.

This work derives the semiclassical Rabi Hamiltonian starting from the electric-dipole approximation, obtains the analytical solution within the rotating-wave approximation, and compares it with the numerical solution of the complete time-dependent model. Particular attention is given to the excitation probability, the effect of detuning, the emergence of Bloch–Siegert oscillations, and the evolution of the atomic state on the Bloch sphere.

The article is organized as follows. Section I derives the atom–field interaction in the electric-dipole approximation. Section II introduces the two-level approximation, Pauli operators, and the Bloch representation. Section III presents the complete semiclassical Rabi model. Section IV introduces the rotating-wave approximation and derives its analytical solution. Section V presents the numerical and analytical results. Section VI discusses the physical interpretation and validity of the approximation, and Section VII summarizes the main conclusions.

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Atom–field interaction in the electric-dipole approximation

Minimal-coupling Hamiltonian

Consider an electron of mass m and charge $-e$. In the presence of scalar and vector potentials, the minimal-coupling Hamiltonian is

$$\hat{H} = \frac{1}{2m} [\hat{\mathbf{p}} + e\mathbf{A}(\hat{\mathbf{r}}, t)]^2 - e\phi(\hat{\mathbf{r}}, t). \quad (1)$$

Here, $\hat{\mathbf{r}}$ is the electron position relative to the nucleus and $\hat{\mathbf{p}}$ is its canonical momentum.

In the Coulomb gauge,

$$\nabla \cdot \mathbf{A}(\mathbf{r}, t) = 0. \quad (2)$$

For a hydrogen-like atom, the static Coulomb potential may be written as

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{Ze}{r}, \quad (3)$$

where Z is the nuclear charge number.

Expanding Eq. (1) and using the Coulomb-gauge condition gives

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} - e\phi(\hat{\mathbf{r}}) + \frac{e}{m} \mathbf{A}(\hat{\mathbf{r}}, t) \cdot \hat{\mathbf{p}} + \frac{e^2}{2m} \mathbf{A}^2(\hat{\mathbf{r}}, t). \quad (4)$$

Long-wavelength approximation

For optical transitions, the wavelength of the driving field is generally much larger than the characteristic atomic size. In particular,

$$\frac{\lambda}{a_0} \sim 10^4 - 10^5, \quad (5)$$

where a_0 is the Bohr radius. The spatial variation of the field over the atomic region can therefore be neglected:

$$\mathbf{A}(\mathbf{r}, t) \approx \mathbf{A}(\mathbf{0}, t). \quad (6)$$

This is known as the long-wavelength approximation.

Within this approximation, the term $\mathbf{A}^2(t)$ is independent of the atomic coordinates. For a prescribed classical field, it contributes only a time-dependent global phase and does not affect atomic populations or other internal observables. It may therefore be omitted, giving

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} - e\phi(\hat{\mathbf{r}}) + \frac{e}{m} \mathbf{A}(t) \cdot \hat{\mathbf{p}}. \quad (7)$$

Transformation to the electric-dipole interaction

Another useful formulation can be obtained through the Göppert–Mayer gauge transformation. Within

the long-wavelength approximation, the transformed Hamiltonian takes the form

$$\hat{H} = \hat{H}_{\text{atom}} + \hat{H}_{\text{int}}, \quad (8)$$

where

$$\hat{H}_{\text{atom}} = \frac{\hat{\mathbf{p}}^2}{2m} - e\phi(\hat{\mathbf{r}}), \quad (9)$$

and

$$\hat{H}_{\text{int}} = -\hat{\mathbf{d}} \cdot \mathcal{E}(t). \quad (10)$$

Here,

$$\hat{\mathbf{d}} = -e\hat{\mathbf{r}} \quad (11)$$

is the electric-dipole operator and

$$\mathcal{E}(t) = -\frac{\partial \mathbf{A}(t)}{\partial t} \quad (12)$$

is the classical electric field. Equation (10) is known as the electric-dipole interaction.

Two-level atom and Bloch representation

Two-level approximation

Atoms possess discrete and generally non-equidistant energy spectra. Their energy levels depend on the electronic structure, orbital angular momentum, spin, relativistic corrections, and internal couplings.

When an atom interacts with a nearly monochromatic laser, the driving frequency can be tuned close to a particular transition frequency. If the remaining transitions are sufficiently far from resonance, the dynamics can be restricted to a ground state $|g\rangle$ and an excited state $|e\rangle$, with energies E_g and E_e , respectively.

The atomic Hilbert space is then approximated by

$$\mathcal{H}_{\text{atom}} \approx \text{span} \{|e\rangle, |g\rangle\}. \quad (13)$$

The free atomic Hamiltonian becomes

$$\hat{H}_{\text{atom}} \approx E_e |e\rangle \langle e| + E_g |g\rangle \langle g|. \quad (14)$$

Defining the transition frequency as

$$\omega_a = \frac{E_e - E_g}{\hbar}, \quad (15)$$

and subtracting the constant energy $(E_e + E_g)\hat{I}/2$, the Hamiltonian can be written as

$$\hat{H}_{\text{atom}} = \frac{\hbar\omega_a}{2} \hat{\sigma}_z. \quad (16)$$

The discarded identity term produces only a global phase and therefore does not affect the physical observables

Pauli and ladder operators

In the ordered basis

$$\{|e\rangle, |g\rangle\}, \quad (17)$$

the Pauli operators are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (18)$$

Equivalently,

$$\hat{\sigma}_x = |e\rangle\langle g| + |g\rangle\langle e|, \quad (19)$$

$$\hat{\sigma}_y = i|e\rangle\langle g| - i|g\rangle\langle e|, \quad (20)$$

$$\hat{\sigma}_z = |e\rangle\langle e| - |g\rangle\langle g|. \quad (21)$$

To effect transition between atomic levels, the ladder operator are introduced as

$$\sigma_+ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \sigma_- = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

They satisfy

$$\hat{\sigma}_+ |g\rangle = |e\rangle, \quad \hat{\sigma}_- |e\rangle = |g\rangle. \quad (22)$$

Therefore, $\hat{\sigma}_+$ excites the atom, whereas $\hat{\sigma}_-$ produces de-excitation.

Pure and mixed states on the Bloch sphere

A general pure state of the two-level atom may be written as

$$|\psi\rangle = c_e |e\rangle + c_g |g\rangle, \quad (23)$$

where

$$|c_e|^2 + |c_g|^2 = 1. \quad (24)$$

After removing an irrelevant global phase, the state can be parametrized by two angular variables:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |e\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right) |g\rangle, \quad (25)$$

where

$$\theta \in [0, \pi], \quad \varphi \in [0, 2\pi). \quad (26)$$

The expectation values of the Pauli operators are

$$\langle \hat{\sigma}_x \rangle = \sin \theta \cos \varphi, \quad (27)$$

$$\langle \hat{\sigma}_y \rangle = \sin \theta \sin \varphi, \quad (28)$$

$$\langle \hat{\sigma}_z \rangle = \cos \theta. \quad (29)$$

These quantities define the Bloch vector

$$\mathbf{R} = (R_x, R_y, R_z), \quad R_k = \text{Tr}(\hat{\rho} \hat{\sigma}_k). \quad (30)$$

The density operator can be expressed as

$$\hat{\rho} = \frac{1}{2} (\hat{I} + \mathbf{R} \cdot \hat{\boldsymbol{\sigma}}). \quad (31)$$

Positivity of the density operator requires

$$|\mathbf{R}| \leq 1. \quad (32)$$

Pure states satisfy $|\mathbf{R}| = 1$ and lie on the surface of the Bloch sphere, whereas mixed states satisfy $|\mathbf{R}| < 1$ and lie inside the sphere.

Semiclassical Rabi model

Consider a linearly polarized monochromatic electric field,

$$\mathcal{E}(t) = \mathcal{E}_0 \cos(\omega_f t + \phi) \mathbf{e}_x, \quad (33)$$

where $\mathcal{E}_0$ is the field amplitude, ω_f is the driving frequency, and ϕ is its phase.

Within the two-level approximation, the dipole interaction contains the matrix elements

$$\mathbf{d}_{eg} = \langle e | \hat{\mathbf{d}} | g \rangle, \quad \mathbf{d}_{ge} = \mathbf{d}_{eg}^*. \quad (34)$$

The phase of the basis states can be chosen such that the relevant dipole matrix element is real. For $\phi = 0$, the Hamiltonian becomes

$$\hat{H}(t) = \frac{\hbar \omega_a}{2} \hat{\sigma}_z + 2\hbar g \cos(\omega_f t) \hat{\sigma}_x, \quad (35)$$

where the coupling parameter is defined as

$$g = -\frac{\mathcal{E}_0}{2\hbar} \mathbf{e}_x \cdot \mathbf{d}_{eg}. \quad (36)$$

Equation (35) is the semiclassical Rabi Hamiltonian. The factor of two in this definition is a convention; it is chosen so that the rotating-frame interaction strength is $\hbar g$.

Transformation to the rotating frame

We introduce the rotating-frame state

$$|\psi(t)\rangle_R = \hat{U}_f^\dagger(t) |\psi(t)\rangle, \quad (37)$$

with

$$\hat{U}_f(t) = \exp\left(-i \frac{\omega_f t}{2} \hat{\sigma}_z\right). \quad (38)$$

The rotating-frame Hamiltonian is

$$\hat{H}_R = \hat{U}_f^\dagger \hat{H} \hat{U}_f - i\hbar \hat{U}_f^\dagger \frac{d\hat{U}_f}{dt}. \quad (39)$$

Defining the detuning

$$\Delta = \omega_a - \omega_f, \quad (40)$$

one obtains

$$\hat{H}_R(t) = \frac{\hbar\Delta}{2}\hat{\sigma}_z + \hbar g(\hat{\sigma}_+ + \hat{\sigma}_-) + \hbar g(\hat{\sigma}_+ e^{2i\omega_f t} + \hat{\sigma}_- e^{-2i\omega_f t}). \quad (41)$$

The time-independent terms are the co-rotating contributions, whereas the terms oscillating at $2\omega_f$ are the counter-rotating contributions.

Writing the rotating-frame state as

$$|\psi(t)\rangle_R = c_e(t)|e\rangle + c_g(t)|g\rangle, \quad (42)$$

the Schrödinger equation gives

$$i\dot{c}_e = \frac{\Delta}{2}c_e + gc_g + gc_g e^{2i\omega_f t}, \quad (43)$$

$$i\dot{c}_g = -\frac{\Delta}{2}c_g + gc_e + gc_e e^{-2i\omega_f t}. \quad (44)$$

These coupled differential equations are solved numerically for the complete semiclassical model.

Rotating-wave approximation

Validity conditions

The rotating-wave approximation is appropriate when the atom is driven near resonance and the coupling is weak compared with the optical frequency:

$$\frac{|\Delta|}{\omega_f} \ll 1, \quad \frac{|g|}{\omega_f} \ll 1. \quad (45)$$

Under these conditions, the counter-rotating terms in Eq. (41) oscillate rapidly compared with the characteristic atomic dynamics and approximately average to zero.

The coefficient equations reduce to

$$i\dot{c}_e = \frac{\Delta}{2}c_e + gc_g, \quad (46)$$

$$i\dot{c}_g = -\frac{\Delta}{2}c_g + gc_e. \quad (47)$$

Effective Hamiltonians

The RWA Hamiltonian in the laboratory frame is

$$\hat{H}_{\text{RWA}}^{\text{lab}}(t) = \frac{\hbar\omega_a}{2}\hat{\sigma}_z + \hbar g(\hat{\sigma}_+ e^{-i\omega_f t} + \hat{\sigma}_- e^{i\omega_f t}). \quad (48)$$

In the rotating frame, it becomes time independent:

$$\hat{H}_{\text{RWA}}^R = \frac{\hbar\Delta}{2}\hat{\sigma}_z + \hbar g\hat{\sigma}_x. \quad (49)$$

Analytical solution

Define

$$\Omega = \sqrt{|g|^2 + \left(\frac{\Delta}{2}\right)^2}. \quad (50)$$

With this convention, 2Ω is the generalized angular frequency of the Bloch-vector precession. The rotating-frame evolution operator is

$$\hat{U}_{\text{RWA}}^R(t) = \begin{pmatrix} \cos(\Omega t) - i\frac{\Delta}{2\Omega}\sin(\Omega t) & -i\frac{g}{\Omega}\sin(\Omega t) \\ -i\frac{g}{\Omega}\sin(\Omega t) & \cos(\Omega t) + i\frac{\Delta}{2\Omega}\sin(\Omega t) \end{pmatrix}. \quad (51)$$

For an atom initially in the ground state,

$$|\psi(0)\rangle = |g\rangle, \quad (52)$$

the excited-state probability is

$$P_e^{\text{RWA}}(t) = \left| \langle e | \hat{U}_{\text{RWA}}^R(t) | g \rangle \right|^2 = \frac{|g|^2}{\Omega^2} \sin^2(\Omega t). \quad (53)$$

Introducing the dimensionless relative detuning

$$\delta = \frac{\Delta}{2g}, \quad (54)$$

Eq. (53) becomes

$$P_e^{\text{RWA}}(t) = \frac{1}{1 + \delta^2} \sin^2(gt\sqrt{1 + \delta^2}). \quad (55)$$

The maximum excitation probability is therefore

$$P_{e,\text{max}}^{\text{RWA}} = \frac{1}{1 + \delta^2}. \quad (56)$$

At resonance, $\Delta = 0$, complete population transfer is possible. Away from resonance, the maximum excited-state probability decreases.

Numerical and analytical results

Numerical method

The complete semiclassical dynamics were calculated by numerically integrating the Schrödinger equation in the rotating frame. The coupling strength g was used as the characteristic frequency scale, and the dimensionless variables

$$\tau = gt, \quad \tilde{\omega}_f = \frac{\omega_f}{g}, \quad \delta = \frac{\Delta}{2g} \quad (57)$$

were introduced. In terms of these variables, the probability amplitudes $c_g(\tau)$ and $c_e(\tau)$ satisfy

$$\frac{dc_g}{d\tau} = -i\delta c_g - i \left(1 + e^{-2i\tilde{\omega}_f\tau}\right) c_e, \quad (58)$$

$$\frac{dc_e}{d\tau} = i\delta c_e - i \left(1 + e^{2i\tilde{\omega}_f\tau}\right) c_g. \quad (59)$$

The equations were integrated using the `solve_ivp` routine from the SciPy library with the explicit Runge–Kutta method of order 5(4), RK45. The relative and absolute error tolerances were set to 10^{-9} and 10^{-11} , respectively. The numerical solution was evaluated at 1000 equally spaced points over the selected integration interval.

Unless otherwise stated, the atom was initially prepared in the ground state,

$$c_g(0) = 1, \quad c_e(0) = 0. \quad (60)$$

For each value of the relative detuning, the default integration interval was chosen as

$$0 \leq \tau \leq \frac{\pi}{\sqrt{1 + \delta^2}}, \quad (61)$$

which corresponds to one complete population oscillation predicted by the rotating-wave approximation. The ground- and excited-state populations were calculated from

$$P_g(\tau) = |c_g(\tau)|^2, \quad P_e(\tau) = |c_e(\tau)|^2. \quad (62)$$

The normalization condition

$$P_g(\tau) + P_e(\tau) = 1 \quad (63)$$

was monitored as a consistency check for the numerical integration.

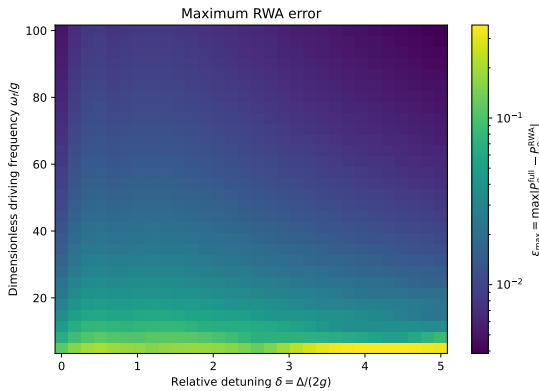


Figure 1: Excited-state probability P_e^{RWA} as a function of the dimensionless time gt for different values of the relative detuning $\delta = \Delta/(2g)$.

Dynamics of the complete semiclassical model

Figure 2 shows the excited-state probability obtained from the complete semiclassical Hamiltonian for several values of the driving frequency and detuning.

Near resonance, the classical field induces periodic transfer of population between the ground and excited states. When the driving frequency is much larger than the coupling strength, $\omega_f \gg g$, the evolution is dominated by a smooth Rabi cycle. As g/ω_f increases, rapid oscillations become superimposed on the slower population dynamics. These fast modulations originate from the counter-rotating terms retained in the complete Hamiltonian.

Rabi oscillations within the RWA

Figure 3 shows the analytical excited-state probability predicted by Eq. (55).

At resonance, the atom periodically reaches the excited state with unit probability. For nonzero detuning, both the oscillation amplitude and the oscillation period are modified. In particular, the oscillation amplitude decreases as $|\delta|$ increases.

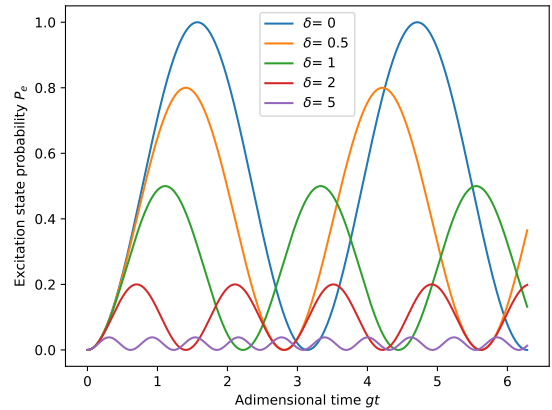


Figure 3: Excited-state probability P_e^{RWA} as a function of the dimensionless time gt for different values of the relative detuning $\delta = \Delta/(2g)$.

Maximum excited-state probability

The analytical maximum excitation probability is given by Eq. (56). It decreases monotonically with the magnitude of the relative detuning.

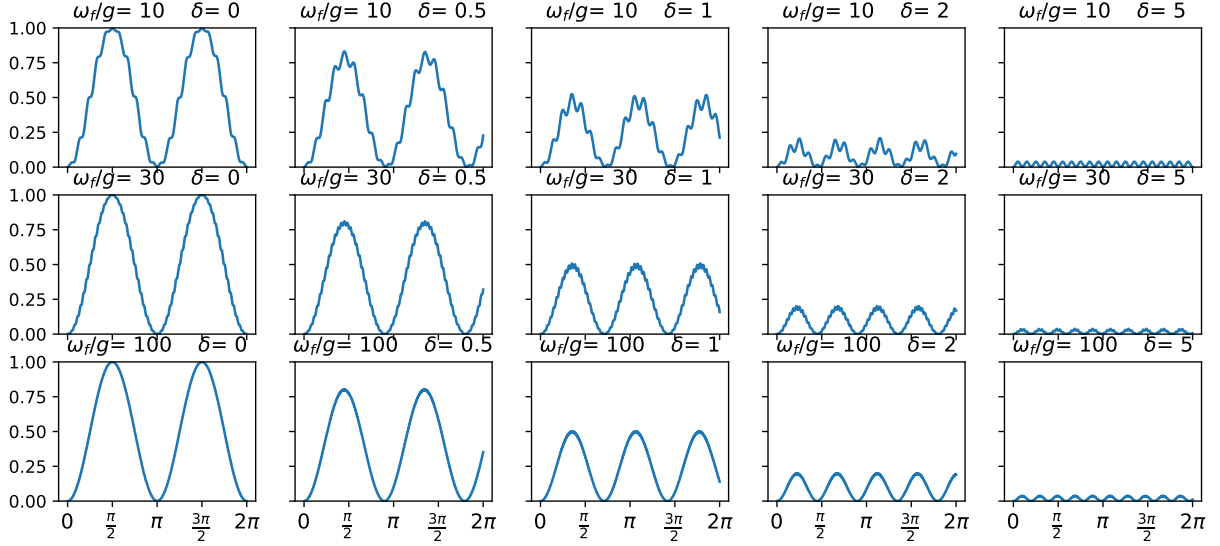


Figure 2: Excited-state probability obtained from the complete semiclassical Rabi Hamiltonian for different values of the driving frequency and detuning. Fast modulations become more pronounced as g/ω_f increases, indicating the progressive breakdown of the rotating-wave approximation.

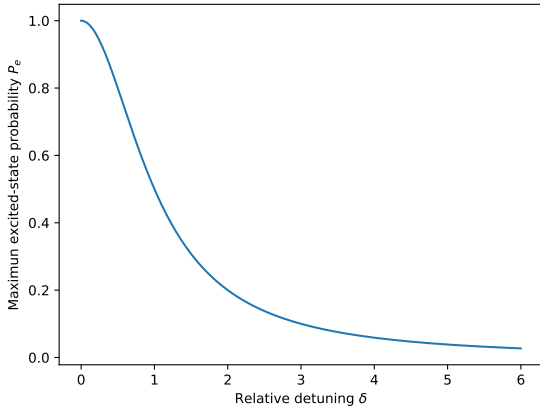


Figure 4: Maximum excited-state probability $P_{e,\max}^{\text{RWA}}$ as a function of the relative detuning δ .

For $|\delta| \gg 1$, the driving frequency is far from the atomic resonance and the transition probability becomes small.

Comparison between the full model and the RWA

A direct measure of the difference between the complete and approximate models is

$$\epsilon_P(t) = \left| P_e^{\text{full}}(t) - P_e^{\text{RWA}}(t) \right|. \quad (64)$$

The maximum population error over an interval $[0, T]$ may be defined as

$$\epsilon_{\max} = \max_{0 \leq t \leq T} \epsilon_P(t). \quad (65)$$

This quantity can be used to determine quantitatively the parameter region in which the rotating-wave approximation remains accurate.

Bloch-sphere dynamics

For the initial ground state, the amplitudes obtained under the RWA are

$$c_e(t) = -i \frac{g}{\Omega} \sin(\Omega t), \quad (66)$$

$$c_g(t) = \cos(\Omega t) + i \frac{\Delta}{2\Omega} \sin(\Omega t). \quad (67)$$

The Bloch-vector components are therefore

$$\langle \hat{\sigma}_x \rangle = \frac{2\delta}{1 + \delta^2} \sin^2(\Omega t), \quad (68)$$

$$\langle \hat{\sigma}_y \rangle = -\frac{1}{\sqrt{1 + \delta^2}} \sin(2\Omega t), \quad (69)$$

$$\langle \hat{\sigma}_z \rangle = -1 + \frac{2}{1 + \delta^2} \sin^2(\Omega t). \quad (70)$$

Thus,

$$\mathbf{R}(t) = (\langle \hat{\sigma}_x \rangle, \langle \hat{\sigma}_y \rangle, \langle \hat{\sigma}_z \rangle). \quad (71)$$

The rotating-frame RWA Hamiltonian can be written as

$$\hat{H}_{\text{RWA}}^R = \frac{\hbar}{2} \mathbf{\Omega}_{\text{eff}} \cdot \hat{\boldsymbol{\sigma}}, \quad (72)$$

where

$$\mathbf{\Omega}_{\text{eff}} = (2g, 0, \Delta). \quad (73)$$

The Bloch vector therefore satisfies

$$\frac{d\mathbf{R}}{dt} = \mathbf{\Omega}_{\text{eff}} \times \mathbf{R}. \quad (74)$$

Consequently, the atomic state precesses around the effective rotation axis

$$\mathbf{n}_{\text{eff}} = \frac{(2g, 0, \Delta)}{\sqrt{4g^2 + \Delta^2}}, \quad (75)$$

with angular frequency

$$\Omega_{\text{Bloch}} = \sqrt{4g^2 + \Delta^2} = 2\Omega. \quad (76)$$

At resonance, the effective rotation axis lies along the x direction. When the detuning is nonzero, the axis tilts toward the z direction, preventing complete population inversion.

Discussion

The analytical solution obtained within the rotating-wave approximation captures the principal features of the driven two-level dynamics. At resonance, the field produces complete and periodic population transfer between the two atomic states. Away from resonance, the maximum excitation probability decreases according to Eq. (56), while the generalized oscillation frequency increases.

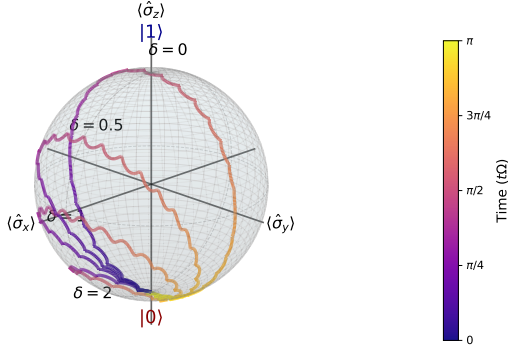
The Bloch-sphere representation provides a geometrical interpretation of these results. The coupling determines the transverse component of the effective rotation axis, whereas the detuning determines its longitudinal component. Complete population inversion is possible at resonance because the effective axis is perpendicular to the initial ground-state Bloch vector. For nonzero detuning, the rotation axis acquires a z component and the trajectory no longer reaches the excited-state pole.

The complete semiclassical Hamiltonian additionally contains counter-rotating terms oscillating at $2\omega_f$. When $g/\omega_f \ll 1$, their net effect is small and the RWA accurately describes the slow atomic dynamics. When this ratio is no longer small, the counter-rotating contributions produce fast micromotion and modify both the oscillation frequency and resonance condition.

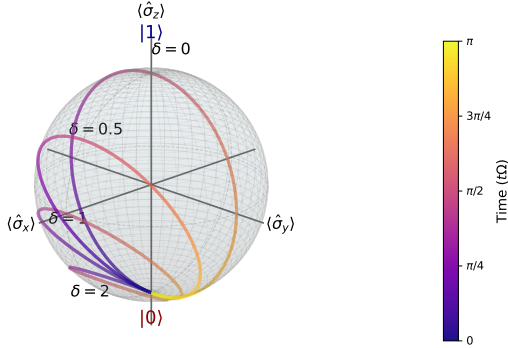
The population error defined in Eq. (64) provides a quantitative criterion for testing the validity of the approximation. Further characterization may be obtained from the state fidelity,

$$F(t) = |\langle \psi_{\text{full}}(t) | \psi_{\text{RWA}}(t) \rangle|^2, \quad (77)$$

or from the distance between the corresponding Bloch vectors,



(a) Bloch-sphere trajectory obtained from the complete semiclassical Hamiltonian.



(b) Bloch-sphere trajectory obtained within the rotating-wave approximation.

Figure 5: Comparison of the atomic Bloch-vector trajectories obtained from the complete model and from the rotating-wave approximation.

$$D_B(t) = |\mathbf{R}_{\text{full}}(t) - \mathbf{R}_{\text{RWA}}(t)|. \quad (78)$$

These quantities are sensitive not only to differences in the populations but also to differences in the relative phase of the atomic amplitudes.